

No. of Functional Features Included in the Solution

Date	13 July 2026
Team ID	RW-2026-01
Project Name	Rising Waters – AI-Based Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

The **Rising Waters** system includes essential features for collecting weather data, predicting flood risks using machine learning, generating alerts, and providing an interactive dashboard for users and disaster management authorities. These features ensure accurate predictions, timely notifications, and an improved disaster response process.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Weather Data Collection	Collects rainfall, temperature, humidity, and river water level data from datasets and APIs.	Data Collection	Done	High
2	Data Preprocessing	Cleans, validates, and prepares historical weather and flood data for model training.	Data Processing	Done	High
3	Flood Prediction Engine	Uses the trained XGBoost model to predict flood risk based on weather parameters.	Machine Learning	Done	High
4	Flood Risk Assessment	Classifies prediction results into different flood risk levels.	Prediction Module	Done	Medium
5	Alert Generation	Generates early warning alerts when flood risk exceeds the predefined threshold.	Alert System	Done	High
6	User Dashboard	Displays prediction results, weather information, and flood alerts through a web interface.	Frontend (Flask + HTML)	Done	Medium
7	Prediction History & Reports	Stores prediction results and provides historical analysis and reports.	Reporting Module	Done	Medium
8	Model Deployment	Deploys the trained machine learning model using the Flask web framework.	Deployment	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Flood Prediction, Data Preprocessing, Weather Data Collection, Alert Generation, Dashboard
Additional / Nice-to-Have Features	Prediction History, Reports, Future Weather API Integration, Cloud Deployment
Features Tested & Verified	8 / 8 Successfully Tested

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Dashboard, Prediction Form, Result Page	Weather Input, Flood Risk Display, Alert Dashboard
2	Backend / Logic	Data Processing, Prediction Engine	XGBoost Prediction, Feature Scaling, Risk Assessment
3	Database / Storage	Dataset & Model Storage	Historical Flood Dataset, Weather Dataset, floods.save, transform.save
4	API / Integration	External Data Sources	Weather API (<i>future enhancement</i>), Rainfall Data Integration
5	Security / Authentication	User Access Management	Login Authentication, Session Management (<i>optional enhancement</i>)

Outcome:

The Rising Waters project successfully implements all planned core functional features, including weather data processing, machine learning-based flood prediction, risk assessment, alert generation, and a user-friendly web dashboard. The completed solution meets the project objectives and provides a scalable foundation for future enhancements such as live weather API integration, cloud deployment, and mobile notifications.